Department of Information Technology
Academic Year 2024 – 2025(Even Semester)

Degree, Semester & Branch: B.Tech-VI & IT
Course Code & Title: CCS356 & Object Oriented Software Engineering
Name of the Faculty member (s): M.Rethinakumari

Innovative Practice Description

- **Unit / Topic: Unit IV / Regression Testing**

Course Outcome: The student will be able to apply regression testing principles by collaborating and presenting on different testing strategies using the Jigsaw Learning methodology.

- **Activity Chosen:** Learning by Teaching
- **Justification:**

Regression Testing is a crucial practice in software maintenance, ensuring that new changes don't negatively affect existing functionality. The Jigsaw Learning Method allows students to explore different types of regression tests and collaborate by presenting their findings to the class. This methodology encourages peer-to-peer learning, enhances teamwork, critical thinking, and problem-solving skills that are key for modern software engineering.

Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**

Ms. Abinaya S G was given the responsibility to prepare and present the topic of **Regression Testing**. She researched key concepts, tools, and best practices related to regression testing.

During the session, **Ms. Abinaya S G** covered the following points:
Importance of Regression Testing**:** Why it is crucial for maintaining software quality.
Types of Regression Testing: Corrective, Progressive, and Retest All.
Tools used for Regression Testing**:** Examples like Selenium and JUnit.
Automation in Regression Testing**:** How automated tests help in faster validation.
Best Practices in Regression Testing**:** Tips and strategies to implement regression testing effectively.

Ms. Ms. Abinaya S G presented the topic in a clear, structured manner, using examples and real-world scenarios to explain the concepts as the figures1 and 2 shows. After her presentation, she engaged the class in a discussion, encouraging questions and clarifying doubts. This helped in further solidifying her understanding and allowed her peers to interact with the topic more deeply.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PSO1	PSO2
CO4	3	3	2	2	3	3	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO8**	**PSO1**	**PSO2**
Justification for correlation	Apply core design principles like cohesion, coupling, and functional independence to build efficient and modular software systems.	Identify and analyze defects in software systems using systematic testing techniques. Emphasize precision and accuracy during defect identification.	Use methods like black-box, white-box, and box, regression testing to ensure functionality and system stability. Validate and verify software systems effectively.	Debug and resolve software issues using advanced tools and techniques like symbolic execution or model checking. Investigate critical system failures efficiently.	Maintain professional ethics while ensuring quality and reliability in software testing. Address issues like data privacy and transparency during testing.	Develop expertise in advanced testing methods to ensure error-free software.	Deliver reliable software solutions by systematically implementing robust testing frameworks. Improve quality assurance with efficient methodologies.

- **Images / Screenshot of the practice:**

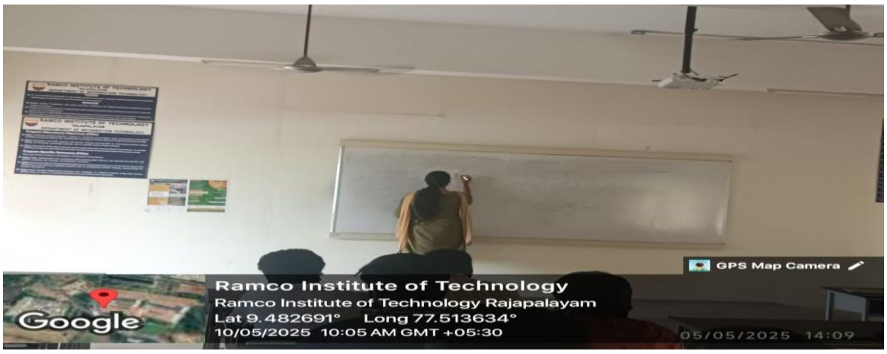
- 

Fig 1: Presentation of S.G.Abinaya

Fig 2: From each group Expert and Member explain their task

Reflective Critique:

❖ **Feedback of Practice from Students and Other Stakeholders:**

Students were able to better remember testing principles and strategies.
Teaching a peer group improved the presenter's self-confidence.
Students found the seminar style more interactive and effective.
Communication and presentation skills were strengthened.

❖ **Benefit of the Practice:**

Students' technical knowledge, confidence, and communication improved.
Peer-to-peer learning created a deeper understanding of Regression Testing.
Helped students develop teaching and leadership qualities.

❖ **Challenges Faced in Implementation:**

Convincing all students to prepare independently was slightly challenging.
Limited time for deeper discussion after the presentation.
Not all students participated equally during the interactive session.

References:

❖ https://www.softwaretestinghelp.com/regression-testing-tutorial/
❖ https://www.geeksforgeeks.org/regression-testing-in-software-testing/

Signature of Faculty Member **HOD**